

The smallest triangle containing two unit circles has area $6 + 4\sqrt{2}$
(a proof via Heron's formula)

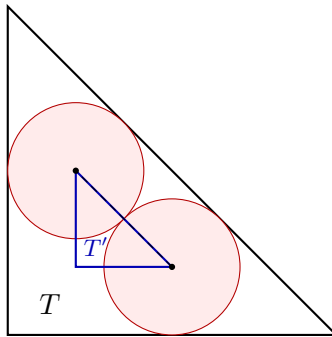
Theorem 1. *Among all triangles containing two interior-disjoint unit circles (radius 1), the minimum area is $6 + 4\sqrt{2} \approx 11.6568542$, attained by the right isosceles triangle with legs $2 + 2\sqrt{2}$ and hypotenuse $4 + 2\sqrt{2}$.*

We use the following steps:

- show there exists an “inner triangle” with longest side at least 2;
- derive the area of the outer triangle in terms of the inner triangle;
- show that the longest side of the inner triangle must be exactly 2;
- show that the inner triangle must be isosceles;
- minimize the area of the outer triangle over all possible short side lengths of the inner triangle.

Step 0. Reduction to the inner triangle

Two unit circles lie in a triangle T with disjoint interiors iff their centers p, q each lie at distance ≥ 1 from all three sides (each circle inside T) and $|p - q| \geq 2$ (circles disjoint). The set of points at distance ≥ 1 from all three sides is the *inner triangle* T' , obtained by moving each side inward by 1. The three inward offset lines stay parallel to the originals and each drops its distance from the incenter of T from R to $R - 1$ (where R is the inradius of T); hence T' has the same incenter and the same angles as T , i.e. it is similar to T with inradius $r' = R - 1$ and similarity ratio $(R - 1)/R$. It is nonempty iff $R > 1$.



The outer triangle T contains the two unit circles; their centers (dots) lie in the inner triangle T' and are distance ≥ 2 apart.

A compact convex set contains two points at distance ≥ 2 iff its diameter is ≥ 2 (the diameter is attained by two of its points), and a triangle's diameter is its longest side. Hence:

$$(\mathbf{R}) \quad T \text{ holds two disjoint unit circles} \iff R > 1 \text{ and the longest side of } T' \text{ is } \geq 2.$$

We minimize $\text{Area}(T)$ subject to $(\mathbf{R})$.

Step 1. Heron form of the objective

Let T have inradius R and semiperimeter σ , so $\text{Area}(T) = \sigma R$. Let s' , A' , r' be the semiperimeter, area, and inradius of T' ; recall the identity $r' = A'/s'$. Since T' is similar to T with ratio $(R-1)/R = r'/(r'+1)$, we have $\sigma = s'(r'+1)/r'$ and $R = r'+1$, so

$$\text{Area}(T) = \sigma R = \frac{s'(r'+1)^2}{r'} = \frac{s'(\frac{A'}{s'}+1)^2}{A'/s'} = \boxed{\frac{(s'+A')^2}{A'}}. \quad (1)$$

Thus we must minimize $(s'+A')^2/A'$ over inner triangles T' whose longest side is ≥ 2 .

Step 2. The longest side of T' equals 2

Fix the shape (angles) of T' and scale it by a factor $t > 0$. The semiperimeter scales linearly and the area quadratically: $s' = S_0 t$, $A' = A_0 t^2$ ($S_0, A_0 > 0$). By (1),

$$\text{Area}(T) = \frac{(S_0 t + A_0 t^2)^2}{A_0 t^2} = \frac{(S_0 + A_0 t)^2}{A_0}, \quad \frac{d}{dt} \text{Area}(T) = \frac{2(S_0 + A_0 t) A_0}{A_0} = 2(S_0 + A_0 t) > 0.$$

So $\text{Area}(T)$ strictly increases with the scale of T' , while **(R)** only bounds the longest side of T' from below by 2. Hence at the optimum the longest side of T' equals exactly 2. Henceforth T' has sides $2, x, y$ with $0 < x, y \leq 2$ and $x + y > 2$ (the side 2 is longest, so $x, y \leq 2$; the triangle inequality gives $x + y > 2$).

Step 3. Lemma: every such T' has $A' < s'$ (equivalently $r' < 1$)

Lemma 1. *If a triangle has longest side 2, then its area is strictly less than its semiperimeter; equivalently its inradius $r' < 1$.*

Proof. Write $u = s' - x$, $v = s' - y$, $w = s' - 2$ for the three Heron terms; all are positive. Two facts: first,

$$u + v = (s' - x) + (s' - y) = 2s' - (x + y) = 2 \quad (\text{since } 2s' = 2 + x + y),$$

so by AM–GM $uv \leq (\frac{u+v}{2})^2 = 1$; and second, $w = s' - 2 < s'$. Heron's formula then gives

$$A'^2 = s' u v w \leq s' \cdot 1 \cdot w = s' w < s' \cdot s' = s'^2,$$

hence $A' < s'$. □

Step 4. Reduction to the isosceles inner triangle

By (1), $\frac{\partial}{\partial A'} \frac{(s'+A')^2}{A'} = 1 - \frac{s'^2}{A'^2}$, which is < 0 by Step 3. Since $s' = (2 + x + y)/2$ depends only on the perimeter, “fix s' ” is the same as “fix the perimeter”; on each such slice $\text{Area}(T)$ is a strictly decreasing function of A' . Therefore, to *minimize* the outer area we *maximize* the inner area A' at each fixed perimeter.

With the perimeter fixed (so $x + y$ fixed and the base fixed at 2), Heron gives $A'^2 = s'(s' - 2)(s' - x)(s' - y)$. The factor $(s' - x) + (s' - y)$ is fixed, so by AM–GM the product $(s' - x)(s' - y)$ is maximized exactly when $s' - x = s' - y$, i.e. $x = y$. (This maximizer is always feasible: $x = y$ gives $x = y = \frac{x+y}{2} \leq 2$, so the side 2 remains longest.) Hence the minimizing T' is isosceles with base 2 and equal legs $x \in (1, 2]$.

Step 5. One-variable minimization

For the isosceles inner triangle with base 2 and legs x , the height is $\sqrt{x^2 - 1}$, so $A' = \sqrt{x^2 - 1}$ and $s' = 1 + x$. By (1),

$$\text{Area}(T) = \frac{(1 + x + \sqrt{x^2 - 1})^2}{\sqrt{x^2 - 1}}, \quad x \in (1, 2].$$

Differentiating,

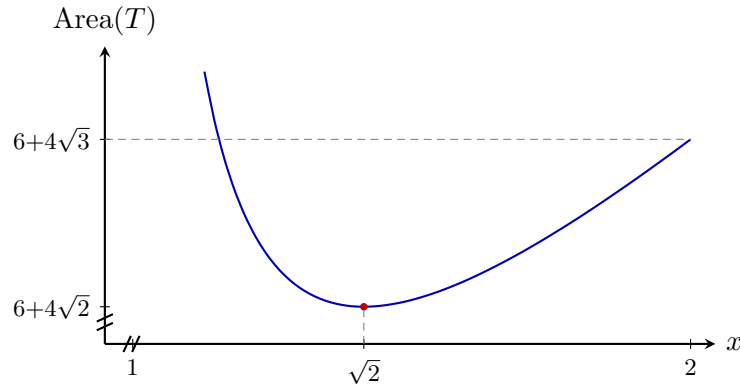
$$\frac{d}{dx} \text{Area}(T) = \frac{2(x^2 + x\sqrt{x^2 - 1} - x - \sqrt{x^2 - 1} - 1)}{(x - 1)^{3/2}\sqrt{x + 1}}.$$

The denominator is positive on $(1, 2]$, so the derivative has the same sign as its numerator. Writing the numerator as $2N(x)$ with

$$N(x) = x^2 + x\sqrt{x^2 - 1} - x - \sqrt{x^2 - 1} - 1 = (x^2 - x - 1) + (x - 1)\sqrt{x^2 - 1},$$

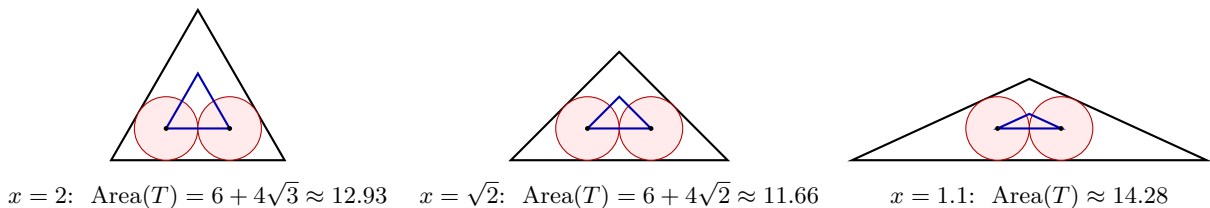
it suffices to determine the sign of $N(x)$ (the factor 2 is positive). Setting $N(x) = 0$ and isolating the radical gives $(x - 1)\sqrt{x^2 - 1} = -(x^2 - x - 1)$; squaring yields $(x - 1)^2(x^2 - 1) = (x^2 - x - 1)^2$, which simplifies to $x^2 - 2 = 0$. The only root in $(1, 2]$ is therefore $x = \sqrt{2}$. Since N is negative for $x < \sqrt{2}$ and positive for $x > \sqrt{2}$, $x = \sqrt{2}$ is the unique minimizer. (At the boundaries $\text{Area}(T) \rightarrow +\infty$ as $x \rightarrow 1^+$, and $\text{Area}(T) = 6 + 4\sqrt{3} \approx 12.93$ at $x = 2$.) At $x = \sqrt{2}$,

$$\text{Area}(T) = \frac{(1 + \sqrt{2} + 1)^2}{1} = (2 + \sqrt{2})^2 = 6 + 4\sqrt{2}.$$



$\text{Area}(T)$ as a function of the inner leg x , with its minimum $6 + 4\sqrt{2}$ at $x = \sqrt{2}$. (The axes are broken near the corner; the plotted window is $x \in [1, 2]$, $\text{Area} \geq 6 + 4\sqrt{2}$.)

The three figures below show the outer triangle T (black), the inner triangle T' (blue) and the two unit circles, for $x = 2$ (equilateral T'), the optimum $x = \sqrt{2}$ (right-isosceles T'), and a thin case $x = 1.1$. The circle centers always sit at the endpoints of T' 's base (length 2); as x moves away from $\sqrt{2}$ in either direction the outer triangle grows.



Step 6. Conclusion

The minimizing inner triangle T' is isosceles with base 2 and legs $\sqrt{2}$; since $2^2 = (\sqrt{2})^2 + (\sqrt{2})^2$ it is right isosceles, with hypotenuse 2 and inradius $r' = A'/s' = 1/(\sqrt{2} + 1) = \sqrt{2} - 1$. Offsetting outward by 1 (ratio $(r' + 1)/r' = \sqrt{2}/(\sqrt{2} - 1) = 2 + \sqrt{2}$) gives the optimal T : right isosceles with legs $\sqrt{2}(2 + \sqrt{2}) = 2 + 2\sqrt{2}$, hypotenuse $2(2 + \sqrt{2}) = 4 + 2\sqrt{2}$, and

$$\text{Area}(T) = \frac{1}{2}(2 + 2\sqrt{2})^2 = 6 + 4\sqrt{2} \approx 11.6568542. \quad \blacksquare$$

The two unit circles sit in the two 45° corners, mutually tangent; their centers are the endpoints of T' 's hypotenuse, exactly distance 2 apart.